

Object oriented Principles

Lab 1

This will contain all your programs for semester one. Now create a new class by clicking that button. Read the information about classes. Now delete the information and start doing your own classes.

Please do question 1 & 4 in the labs. You can do question 2 and 3 either at home or in the labs.

1. Write an application using an **int** and **String** variable, a **while** loop and **if** tests to output the following song to the screen:

space

99 oranges fell on the floor

99 oranges fell

space

Pass me an orange to hit the wall

I really like the smell

space

98 oranges fell on the floor

space

Note: we have only one orange when we reach the end.

The final line is "No more oranges on the floor"

2. Write a java program that request the end-user their name, age, and height.
All details including asterisks must be printed exactly the way it is shown below.

```
*****  
Your name is ---  
Your height is ---  
Your age is ---  
*****
```

3. Write a program that request from end-user their name, the city they live and their age and display them as follows:

```
*****  
My name is Deirdre  
I live in Cork  
I am nearly 93 years old  
*****
```

4. Using functions write a menu program for David that will help him with the following problems
- David wants you to print a payslip telling him his gross pay. He will supply his hourly rate of pay, the number of standard-rate hours he has worked and the number of overtime hours. His overtime rate is “time and a half”, that is, 1.5 times his standard rate. Please display details for him.
 - When David does retire, he will be entitled to an annual pension of one sixtieth of his salary for each complete years’ service with his company. If Charlie supplies his annual salary and the number of years he has served, tell him what his annual pension will be.